

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Quadratics

Turn every quadratic into a form you can read: roots, vertex, sign, or graph.

QUADRATICS · P1 · 02

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA11 website answer bank. The topic reduces to these exam moves.

01 ROOT: Factor To Solve

TRIGGER *the quadratic is equal to zero*

02 SQUARE: Complete The Square

TRIGGER *minimum, maximum, turning point, vertex*

03 FORM: Formula And Discriminant

TRIGGER *not factorisable, exact roots, number of roots*

04 GRAPH: Read The Parabola

TRIGGER *sketch, intersections, above or below the axis*

BANK EVIDENCE Local pages: 35 problem, 72 answer, 20 notes. Website primary entries: 16.

01 ROOT: Factor To Solve

TRIGGER *the quadratic is equal to zero*

BECOMES A product where one factor must be zero.

FIRST LINE TO WRITE

$$ax^2 + bx + c = 0$$

SIMPLEST STRATEGY

- 1 Rearrange so one side is zero.
- 2 Open or factor the expression.
- 3 One bracket equal to zero.
- 4 Take the roots and check them.

WORKED MODEL

$$x^2 - 5x + 6 = 0 \Rightarrow (x - 2)(x - 3) = 0, \quad x = 2, 3.$$

HARD VARIANTS

- 1 Hidden quadratic: if you see x^4, x^2 or a^{2x}, a^x , let one repeated expression be a new letter first.
- 2 Parameter question: factor normally, then use the required number of roots or repeated-root condition.
- 3 After solving, return to the original variable and reject values that do not fit the substitution.

— BOTTOM LINE

A factorised quadratic is solved by setting each factor equal to zero.

02 SQUARE: Complete The Square

TRIGGER *minimum, maximum, turning point, vertex*

BECOMES A quadratic written as one square plus a constant.

FIRST LINE TO WRITE

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2$$

SIMPLEST STRATEGY

- 1 Start with the $x^2 + bx$ part.
- 2 Quarter the coefficient effect by halving b .
- 3 Use the bracket square.
- 4 Adjust the constant.
- 5 Read the vertex.
- 6 Extract roots if required.

WORKED MODEL

$$x^2 + 6x + 5 = (x + 3)^2 - 9 + 5 = (x + 3)^2 - 4.$$

HARD VARIANTS

- 1 If $a \neq 1$, factor a from the x^2, x terms before completing the square.
- 2 For range questions, read the vertex and decide whether the parabola opens up or down.
- 3 For tangent/minimum wording, complete the square first, then use the vertex condition.

— BOTTOM LINE

The number inside the bracket gives the vertex after changing sign.

03 FORM: Formula And Discriminant

TRIGGER *not factorisable, exact roots, number of roots*

BECOMES A discriminant check or formula substitution.

FIRST LINE TO WRITE

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

SIMPLEST STRATEGY

- 1 Find a, b, c .
- 2 Obtain $b^2 - 4ac$.
- 3 Roots use the formula.
- 4 Meaning: positive, zero, or negative discriminant.

WORKED MODEL

$$2x^2 - 3x - 1 = 0 \Rightarrow x = \frac{3 \pm \sqrt{9 + 8}}{4} = \frac{3 \pm \sqrt{17}}{4}.$$

HARD VARIANTS

- 1 When coefficients contain letters, keep a, b, c symbolic and use the discriminant condition.
- 2 For equal roots use $b^2 - 4ac = 0$; for no real roots use $b^2 - 4ac < 0$.
- 3 Simplify exact roots only after the formula is correct.

— BOTTOM LINE

The discriminant tells you the type of roots before solving.

04 GRAPH: Read The Parabola

TRIGGER *sketch, intersections, above or below the axis*

BECOMES A curve whose roots and vertex control the answer.

FIRST LINE TO WRITE

$$y = a(x - r_1)(x - r_2)$$

SIMPLEST STRATEGY

- 1 Get the roots or intercepts.
- 2 Read the direction from a .
- 3 Axis of symmetry is halfway between roots.
- 4 Plot the vertex/intercepts.
- 5 Highlight the required region.

WORKED MODEL

$$y = (x - 1)(x - 5)$$

has roots 1, 5, axis $x = 3$, and opens upwards.

HARD VARIANTS

- 1 For inequalities, roots are boundaries; the graph decides inside or outside.
- 2 If the curve is transformed, find the new vertex/intercepts before shading.
- 3 Check open or closed endpoints when the sign is strict or inclusive.

— BOTTOM LINE

For inequalities, the graph tells you which interval is wanted.

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
the quadratic is equal to zero	$ax^2 + bx + c = 0$
minimum, maximum, turning point, vertex	$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2$
not factorisable, exact roots, number of roots	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
sketch, intersections, above or below the axis	$y = a(x - r_1)(x - r_2)$